Environment Setup

SQL Server, SSMS and the course databases, on Windows and Mac

The guide follows the Environment Setup lesson. It adds what the lesson could not show: the newer screens, the other ways to run SQL Server, and the fixes for the common errors.

1 Choose your setup

Pick the path for your computer and read only its pages. Each path starts on its own page and ends with a green box: when you reach it, your setup is done.

Your computer	Path	Same screens as the videos	Cost	Start
Windows	**SQL Server Express + SSMS**	yes	no cost	page 3
Mac	**Docker** and Visual Studio Code	no, but the same SQL	no cost	page 6
Mac	**Parallels** with Windows	yes, SSMS inside Windows	about 100 dollars a year, 14-day trial	page 8
Both	**MySQL or PostgreSQL** instead of SQL Server	no, some queries differ	no cost	page 9

My advice

- **On Windows:** the first path. It is the one in the videos.
- **On a Mac:** Docker. It costs nothing, it is light, and the SQL is the same.
- **On a Mac, with SSMS exactly as in the videos:** Parallels, if your Mac has 16 GB of memory.

For everyone

- What you get in the materials: page 2
- The projects and AdventureWorks: page 11
- When something breaks: page 12

2 What you get in the materials

Download the course materials once. You get this folder:

```
sql-ultimate-course/
├── 00_setup/              this guide, and the one-click Docker start for Mac
├── 01_datasets/           the two course databases
│   ├── 00_sql-server/     for SQL Server, the one in the videos
│   ├── 01_mysql/          the same databases for MySQL
│   ├── 02_postgres/       the same databases for PostgreSQL
│   └── 03_csv-files/      SalesDB as CSV files, for the import lesson
├── 02_docs/               the slides of every chapter, as PDF
└── 03_scripts/            every query of the course, one file per chapter
```

The course uses two databases:

Database	Used in
`MyDatabase`	SELECT, DDL, DML, Filtering Data, Joins, String Functions, Aggregate Functions
`SalesDB`	everything else, from Set Operators to AI and SQL

Every script in `03_scripts/` that works on a database starts with a `USE` line that names it.

Windows

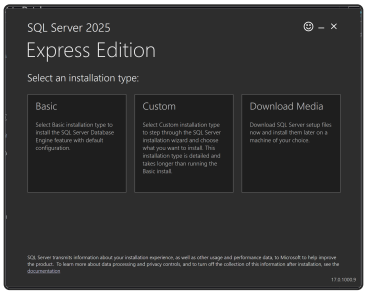
3 Install SQL Server and SSMS

SQL Server Express costs nothing, and it is all you need. The videos show SQL Server 2022. The download page now offers 2025, and it works the same.

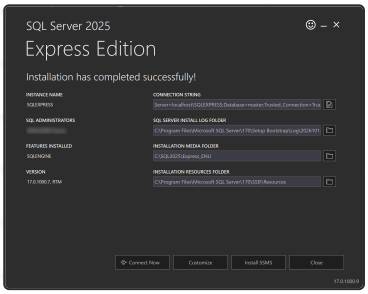
- Download SQL Server Express:
  - Open the SQL Server downloads page.
  - Scroll to the **Express** card.
  - Click **Download now**.
- Run the downloaded file:
  - Choose **Basic**.
  - Accept the licence and click **Install**.
- The last screen shows your server name: `localhost\SQLEXPRESS` . Keep it, you need it in a minute. Your Windows account is now the administrator of this SQL Server.
- Download SQL Server Management Studio (SSMS) and install it:
  - Keep the default options.
  - The installer may offer extra components. You need none of them.



Step 1: the Express card.



Step 2: choose Basic.

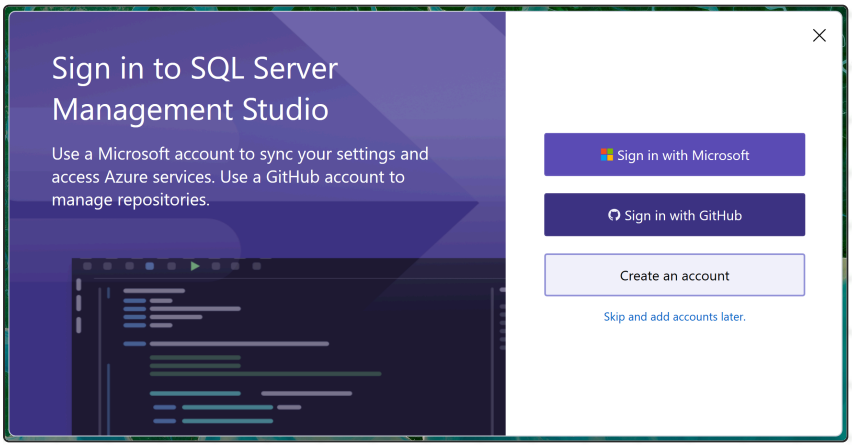


Step 3: the server name.

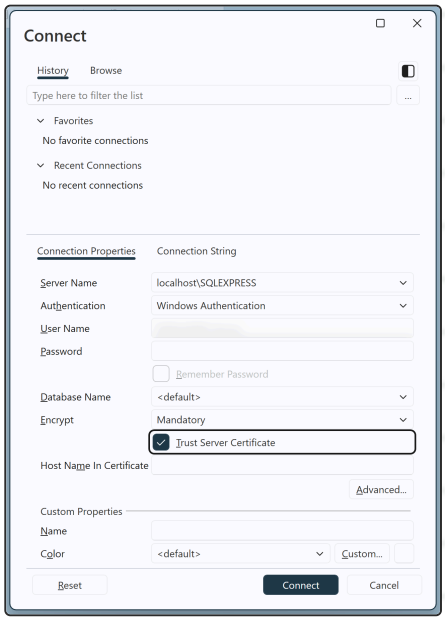
Connect for the first time

SSMS 21 and newer look different from the videos.

- At the first start, SSMS asks you to sign in. You do not need an account: click **Skip and add accounts later** under the buttons.
- In the **Connect** window, fill in three fields:
  - **Server name:** `localhost\SQLEXPRESS`. If the field is empty, that is normal.
  - **Authentication:** Windows Authentication.
  - **Trust Server Certificate:** ticked. Without it you get a certificate error.
- Click **Connect**. Your server appears on the left, in the Object Explorer.



Step 1: the skip link under the buttons.



Step 2: the Connect window, filled in.

4 Create the databases

Pick one of the two ways. Way 1 is the easiest and works on every version of SQL Server.

Way 1: run the scripts

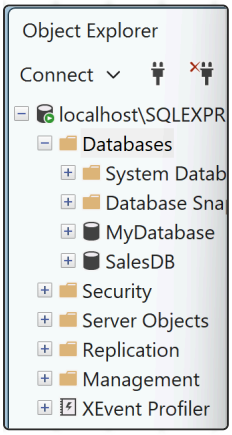
- In SSMS, open the first script:
  - Click **File > Open > File**.
  - Pick `01_datasets/00_sql-server/init-sqlserver-mydatabase.sql`.
- Click **Execute**, or press F5. Refresh the Databases folder: `MyDatabase` is there.
- Do the same with `init-sqlserver-salesdb.sql`. Refresh again: `SalesDB` is there.

Use only the scripts in `00_sql-server/`. The scripts in `01_mysql/` and `02_postgres/` are for other engines, and SSMS cannot run them.

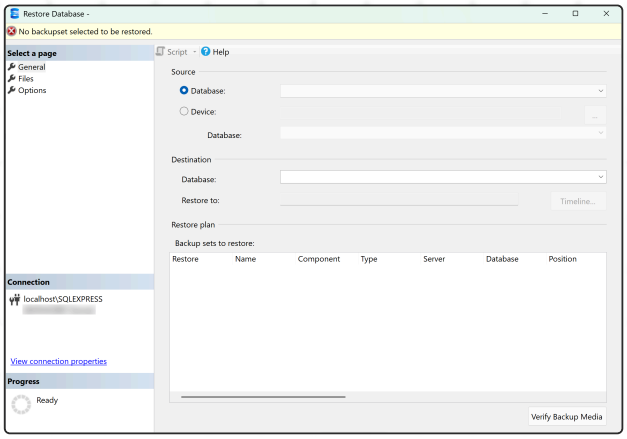
OR

Way 2: restore the backups

- Copy `MyDatabase.bak` and `SalesDB.bak` into the Backup folder of your SQL Server. SQL Server cannot read your Downloads folder, so this copy matters.
  - **SQL Server 2025:** `C:\Program Files\Microsoft SQL Server\MSSQL17.SQLEXPRESS\MSSQL\Backup`
  - **SQL Server 2022:** the same path with `MSSQL16`
- In the Object Explorer, right-click **Databases > Restore Database**.
- In the Restore window:
  - Choose **Device** and click the three dots.
  - Click **Add** and pick the backup file.
  - Click **OK** twice.



The result.



Way 2: Device, the three dots, Add.

> One rule to know: a backup restores on the same SQL Server version or a newer one, never on an older one. The course backups come from SQL Server 2022. On SQL Server 2019 or older, use way 1.

The third way in the lesson

The lesson also imports the CSV files with **Tasks > Import Flat File**, as a demo. Watch it, but build the databases with way 1 or way 2: a CSV file cannot tell an empty text from NULL, and the NULL lessons need that difference.

> **Done.** SQL Server, SSMS and both course databases are ready. Open a script from `03_scripts` and start the course.

Mac with Docker

5 Start SQL Server in Docker

SQL Server does not install on macOS, and SSMS is Windows only. With Docker, SQL Server runs in a small container on your Mac, and you write your queries in Visual Studio Code. The SQL is the same.

Install, once

- Install Docker Desktop, the Apple chip version. Start it and leave it running.
- Open the Docker Desktop settings, **General**. SQL Server ships for Intel chips only, and these two settings let your Mac run it:
  - **Virtual Machine Manager:** Apple Virtualization framework.
  - **Use Rosetta for x86_64/amd64 emulation on Apple Silicon:** on.
- Install the tool you write your queries in:
  - Visual Studio Code.
  - Inside it, the extension SQL Server (mssql) by Microsoft. It is Microsoft's tool for the Mac.

Way 1: one click

- In the course materials, open `00_setup/mac-docker` and double-click `start.command`.
  - **macOS blocks it?** Open System Settings > Privacy & Security, scroll down, click **Open Anyway**, then double-click again. Only the first time.
- Wait a minute. The first start downloads SQL Server and creates `MyDatabase` and `SalesDB` from the course scripts.
- Next time, double-click the same file again.

OR

Way 2: by hand

This command starts an empty SQL Server 2022. Choose your own password: at least 8 characters with upper case, lower case, a number and a symbol.

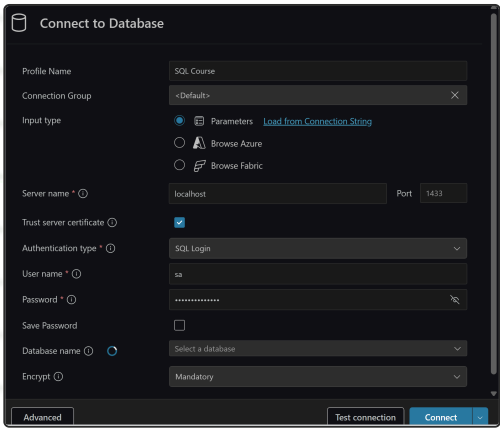
```
docker run -e "ACCEPT_EULA=Y" -e "MSSQL_SA_PASSWORD=YourStrong!Passw0rd" \
  -p 1433:1433 --name sql --platform linux/amd64 \
  -d mcr.microsoft.com/mssql/server:2022-latest
```

After you connect, run `init-sqlserver-mydatabase.sql` and `init-sqlserver-salesdb.sql` from Visual Studio Code. Microsoft's own guide: SQL Server in a Docker container.

Connect

In Visual Studio Code, open the SQL Server extension on the left and add a connection:

- **Server name:** `localhost`
- **Port:** `1433`
- **Trust server certificate:** ticked
- **Authentication type:** SQL Login
- **User name:** `sa`
- **Password:** `SqlServer!2026`, or your own from way 2



The connection form, filled in.

Done. SQL Server runs in Docker with both course databases. Open a script from `03_scripts` and start the course.

Mac with Parallels

6 Install Windows in Parallels

This path gives you SSMS exactly as in the videos, because you run Windows on your Mac.

- Install Parallels Desktop. It downloads and installs Windows 11 for you in a few clicks.
  - **Parallels:** about 100 dollars a year, with a 14-day trial.
  - **Windows:** needs its own licence.
- Give the virtual machine memory:
  - **At least:** 6 GB for Windows.
  - **Your Mac:** 16 GB is the comfortable minimum.
- Start Windows in Parallels. From here on, you set up SQL Server like any Windows user.

SQL Server Express installs and runs on the ARM version of Windows 11, even though Microsoft does not list it as supported. Tested with SQL Server 2025 Express and SSMS 22 in Parallels on an Apple chip.

Next: inside Windows, follow the Windows path from page 3.

MySQL or PostgreSQL

7 Install and load the databases

The course is recorded on SQL Server, and that is the engine I recommend: every query in the videos runs as shown. If you still want MySQL or PostgreSQL, pick one, load the two databases, and keep the table of differences at hand.

MySQL

- Install two things. On Windows, the MySQL Installer brings both:
  - MySQL Community Server, the database.
  - MySQL Workbench, the query tool.
- In Workbench, load the first database:
  - Click **File > Open SQL Script**.
  - Pick `01_datasets/01_mysql/init-mysql-mydatabase.sql`.
  - Run it with the lightning icon.
- The same with `init-mysql-salesdb.sql`.

OR

PostgreSQL

- Install PostgreSQL and a query tool:
  - **Windows:** PostgreSQL with the EDB installer, which brings pgAdmin.
  - **Mac:** Postgres.app, with pgAdmin or DBeaver as the query tool.
- Create the first database:
  - In the query tool, connect to the database `postgres`.
  - Open `01_datasets/02_postgres/init-postgres-mydatabase.sql`.
  - Run only its first two lines, `DROP DATABASE` and `CREATE DATABASE`.
- Connect to the new database, `MyDatabase`, and run the rest of the script.
- The same with `init-postgres-salesdb.sql`. Its database is `salesdb`.

Both engines keep the table names in lower case: `customers`, `orders`, `orders_archive` in MySQL, and `sales.customers` in PostgreSQL.

8 What is different

Most of the course runs unchanged: SELECT, filtering, joins, set operators, CASE, aggregations, window functions, subqueries, CTEs, views. These are the parts that differ, and how to write them on the other engines.

In the course (SQL Server)	MySQL	PostgreSQL
`SELECT TOP 3 * FROM t`	`SELECT * FROM t LIMIT 3`	`SELECT * FROM t LIMIT 3`
`GETDATE()`	`NOW()`	`NOW()`
`DATETRUNC(month, d)`	`DATE_FORMAT(d, '%Y-%m-01')`	`DATE_TRUNC('month', d)`
`DATENAME(month, d)`	`MONTHNAME(d)`	`TO_CHAR(d, 'Month')`
`DATEPART(year, d)` , `YEAR(d)`	`YEAR(d)`	`EXTRACT(YEAR FROM d)`
`EOMONTH(d)`	`LAST_DAY(d)`	`(DATE_TRUNC('month', d) + INTERVAL '1 month - 1 day')::date`
`DATEADD(day, 7, d)`	`DATE_ADD(d, INTERVAL 7 DAY)`	`d + INTERVAL '7 days'`
`DATEDIFF(day, a, b)`	`DATEDIFF(b, a)` , days only	`b - a` for dates
`FORMAT(d, 'yyyy-MM-dd')`	`DATE_FORMAT(d, '%Y-%m-%d')`	`TO_CHAR(d, 'YYYY-MM-DD')`
`CONVERT(DATE, x)`	`CAST(x AS DATE)`	`CAST(x AS DATE)` or `x::date`
`ISNULL(a, b)`	`IFNULL(a, b)` or `COALESCE(a, b)`	`COALESCE(a, b)`
`IIF(cond, a, b)`	`IF(cond, a, b)`	`CASE WHEN cond THEN a ELSE b END`
`LEN(s)`	`CHAR_LENGTH(s)`	`LENGTH(s)`
`CHARINDEX(x, s)`	`LOCATE(x, s)`	`POSITION(x IN s)`
`'a' + 'b'` for text	`CONCAT('a', 'b')`	`'a' || 'b'`
`[Column Name]`	`` `Column Name` ``	`"Column Name"`
`Sales.Customers`	`customers` , MySQL has no schemas inside a database	`sales.customers`
`#temp` tables	`CREATE TEMPORARY TABLE`	`CREATE TEMP TABLE`
`SELECT ... INTO new_table`	`CREATE TABLE new_table AS SELECT ...`	`CREATE TABLE new_table AS SELECT ...`
`BULK INSERT`	`LOAD DATA INFILE`	`COPY ... FROM`

The chapters on stored procedures, triggers, indexes, partitions, execution plans and performance tips are SQL Server chapters. The ideas carry over, the syntax does not. When a query from those chapters fails on another engine, that is the reason, not a mistake in your setup.

Done. Both course databases are loaded. Open a script from `03_scripts` and start the course.

9 The projects

The data warehouse project and the data analytics project come as their own packages. Each one has a `00_setup` folder with its own guide and its own one-click Docker start, so open that guide when you get there.

10 AdventureWorks

The index chapter uses Microsoft's sample database `AdventureWorksDW2022`. It is not part of the course materials, so get it from Microsoft:

- Open the AdventureWorks sample databases page at Microsoft Learn.
- Download the right file:
  - **Take:** `AdventureWorksDW2022.bak`, the data warehouse with `FactInternetSales`.
  - **Not:** `AdventureWorks2022.bak`, the operational database. It does not have that table.
- Restore it like the course backups, section 4, way 2.

The 2022 files need SQL Server 2022 or newer. On SQL Server 2019, take the 2019 files from the same page: the data is the same.

11 When something breaks

Find your message in the first column. The fix is in the last one.

Install and connect

What you see	Why	What to do
SSMS asks you to sign in at the first start	optional account	Click **Skip and add accounts later**
The server name field is empty	normal	Type `localhost\SQLEXPRESS`
Error 26, "Error Locating Server/Instance Specified"	SQL Server is not running, or the name is wrong	Open **Services**, start "SQL Server (SQLEXPRESS)". If you installed the Developer edition, the server name is just `localhost`
"The certificate chain was issued by an authority that is not trusted"	SSMS encrypts the connection	Tick **Trust Server Certificate**
Installer: "Wait on the Database Engine recovery handle failed"	SQL Server could not start during setup	Uninstall SQL Server, restart Windows, install again. If it fails again, the real reason is in the ERRORLOG file under `C:\Program Files\Microsoft SQL Server\MSSQL17.SQLEXPRESS\MSSQL\Log`
SQL Server 2025 setup stops at the rules check	installer problem on some machines	Install SQL Server 2022 Express instead. Everything here runs on it
Your laptop does not let you install anything	no administrator rights	SQL Server needs them. Ask IT, or use a personal computer

Databases and files

What you see	Why	What to do
Msg 262, "CREATE DATABASE permission denied in database 'master'"	your Windows login is not an administrator of this SQL Server. Someone else installed it, often IT	Log in with the Windows account that installed SQL Server, or ask an administrator to add your login to the **sysadmin** role (Security > Logins > Properties > Server Roles)
Msg 911, "Database 'MyDatabase' does not exist", or many syntax errors with backticks	you ran the MySQL or PostgreSQL script in SSMS	Use the scripts in `01_datasets/00_sql-server/`
The restore window does not show the .bak file, or "Access is denied"	SQL Server cannot read Downloads	Copy the file into the Backup folder, section 4, way 2
Msg 3169, the backup "is incompatible with this server"	your SQL Server is older than 2022	Run the scripts, way 1, or install SQL Server 2022 or 2025 Express
Msg 5133, "Directory lookup for the file … failed", in the partitions lesson	the script uses the data folder of SQL Server 2022 Express	Find your data folder with `SELECT physical_name FROM sys.database_files` and use it in the script. In Docker it is `/var/opt/mssql/data/`
"'DATETRUNC' is not a recognized built-in function name"	SQL Server older than 2022	Install SQL Server 2022 or newer. The course uses functions from 2022

What you see	Why	What to do
A .sql file opens in Notepad, or the .bak shows as text in the browser	the file is not linked to SSMS	Open SSMS first, then **File > Open > File**. For downloads, right-click the link and choose **Save link as**

Docker

What you see	Why	What to do
macOS: "start.command" cannot be opened, or "Apple could not verify" it	the file comes from the internet and is not signed	System Settings > Privacy & Security, click **Open Anyway**, then double-click again
"Cannot connect to the Docker daemon"	Docker Desktop is not running	Start Docker Desktop, wait for the whale icon, try again
"no matching manifest for linux/arm64"	SQL Server is built for Intel chips	Turn on Rosetta in Docker Desktop, section 5. By hand, add `--platform linux/amd64` to the command
"port is already allocated"	another SQL Server listens on 1433	Stop it, or change the first `1433` in `docker-compose.yml` to `1434` and connect to port `1434`
The container stops right after the start	the password breaks the rules	At least 8 characters with upper case, lower case, a number and a symbol
"SQL Server did not answer" from start.command	SQL Server could not start, most often Rosetta is off	Docker Desktop settings, **General**, turn on Rosetta, section 5. The container's log in Docker Desktop names the reason

If you are still stuck, write in the Q&A with the exact error message and a screenshot of the window. Then I can see what happened.